COMPREHENSIVE ROOT CAUSE ANALYSIS (RCA) REPORT

Proxmox VE Cluster & Ceph Storage Resilience Testing Incidents

Document Metadata	Details
Cluster Name	`cisai-pve` (Proxmox VE 8.x Enterprise Cluster)
Cluster Nodes	`prox22` (`10.31.6.172`), `prox23` (`10.31.6.173`), `prox24` (`10.31.6.174`)
Incident Classification	Physical Fault Injection & Post-Testing Cluster Transition Artifacts
Date & Time	September 24, 2026
Environment	Production Proxmox HA 3-Node Cluster (12x 20TB SAS Ceph OSDs + 3x 127 TiB ZFS Pools)
Impact Level	Zero Data Loss (100% Data Integrity Preserved) / Transient Management UI Degradation
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1. Executive Summary

During physical datacenter fault-tolerance testing (including 10G interconnect cable pullout tests, SAS drive hot-swaps, and node power-cycle failover scenarios), four distinct phenomena were observed:

1. **Ceph Storage Recovery Delay & `HEALTH_WARN` State:** Ceph OSDs took time to recover, with some OSDs reporting `(5) Input/output error` and placement groups (PGs) remaining degraded.

2. **Web GUI "?" Question Marks & Blue Wrench Icons:** `prox23` and `prox24` appeared with question mark icons or wrench icons in the Proxmox Web GUI sidebar tree, with their resources listed as `unknown`.

3. **VM Migration & Freeze Locks:** Virtual Machines (9001–9004) temporarily displayed `migrate` or `freeze` state in HA Manager.

4. **Systemd "Transaction is Destructive" Service Start Failures:** Initial attempts to restart `pvestatd` and `pvedaemon` on member nodes returned `pam_nologin` and transaction conflict warnings.

This RCA document provides the **deep technical explanations** for why each of these behaviors occurred, why they are expected consequences of specific Linux kernel / Proxmox HA subsystems during physical pull testing, and how they were completely resolved to bring the cluster to **100%** `HEALTH_OK`.

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2. Technical Root Causes & Mechanisms



Root Cause 1: Why Ceph Storage Takes Time & Throws I/O Errors During Drive Hot-Pulls

Mechanism:

SCSI Dynamic Minor Number Assignment: When an enterprise SAS/SATA drive is physically unlatched from the backplane and re-inserted, the Linux SCSI mid-layer detects a removal of the device handle and assigns the *next sequential SCSI letter* upon re-insertion (e.g., the drive originally mapped as `/dev/sdb` with Major:Minor `8:16` is re-enumerated as `/dev/sdn` with Major:Minor `8:208`).

LVM Device-Mapper Desync: Ceph Bluestore uses LVM Logical Volumes (`/dev/ceph-/osd-block-`). When the physical drive returns as `/dev/sdn`, the in-memory Linux Device-Mapper table (`/dev/dm-7`) still points to the dead major:minor target `8:16`.

Direct I/O Failure: When `ceph-volume` or `ceph-osd` attempts to read the Bluestore superblock via `direct_aligned_read`, the kernel returns `EIO (Error 5: Input/output error)`.

Ceph Heartbeat & Recovery Timers: Ceph OSDs exchange heartbeat pings every 6 seconds. If an OSD misses heartbeats for `osd_heartbeat_grace` (20 seconds), the Ceph MON leader marks the OSD as `down`. Upon drive re-insertion and service startup, Ceph must:

1. Peer placement groups (PGs).

2. Replay Bluestore write-ahead logs (WAL).

3. Reconcile degraded or misplaced objects across surviving replicas.

This intentional reconciliation phase causes Ceph to transition from `undersized+degraded` -> `active+clean`, taking between 30 to 90 seconds depending on pending PG logs.

Root Cause 2: Why the Proxmox Web GUI Showed "?" and Blue Wrench Icons

Mechanism:

HA Maintenance Mode Flag: During node failover and recovery testing, `prox23` and `prox24` entered HA Maintenance mode (`lrm prox23 (maintenance mode)`).

UI Representation:

* **Blue Wrench Icon:** Proxmox Web UI renders a blue wrench icon next to any node that has an active `maintenance` state in `/etc/pve/ha/manager_status`.

* **Question Mark `?` Icon & "Unknown" Status:** When a node enters maintenance mode or undergoes a reboot, the `pvestatd` (Proxmox VE Cluster Metric Daemon) on that node is suspended. The cluster resource API (`/cluster/resources`) stops receiving live CPU, memory, and storage metrics for that node, marking its status as `status: "unknown"` (which ExtJS displays as a `?`).

Resolution: Disabling maintenance mode (`ha-manager node-maintenance disable`) and restarting `pvestatd` immediately resumes metric broadcasting, turning all node icons to solid **green checkmarks**.

Root Cause 3: Why Initial Systemctl Start Failed with "Transaction is Destructive"

Mechanism:

Pending Server Reboot Sequence: When a reboot was initiated on `prox23` and `prox24`, systemd created `/run/nologin` (PAM login lock) and queued `stop` jobs for foundational targets (`systemd-modules-load.service` , `remote-fs.target` , `dev-pve-swap.swap`).

Transaction Conflict: Executing `systemctl start pvestatd` via SSH while systemd already had an active `stop` job queued in its transaction pipeline caused systemd to reject the start request with:

```
Failed to start pvestatd.service: Transaction for pvestatd.service/start is destructive

(systemd-modules-load.service has 'stop' job queued, but 'start' is included in transaction).
```

Resolution: Allowing the enterprise servers to finish their full hardware reboot cycle (BIOS POST, SAS HBA discovery, systemd target initialization) allowed systemd to start all cluster daemons in the clean, dependency-ordered sequence.

Root Cause 4: Why VMs Displayed "Migrate" and "Freeze" States

Mechanism:

Proxmox Cluster Resource Manager (CRM) Protection: When `prox23` and `prox24` became unreachable during the 10G cable pullout and reboot tests, the Proxmox HA Manager's watchdog mechanism froze local resources (`freeze` state) to prevent split-brain execution.

Migration Locks: VMs scheduled for failover had `lock: migrate` written to their configuration files in `/etc/pve/nodes//qemu-server/.conf`.

Resolution: Once quorum was restored (3/3 votes), the CRM leader (`prox22`) released the migration locks and resumed VMs into their configured running states.

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3. Incident Timeline & Telemetry Summary

Time (IST)	Event / Test Phase	System Telemetry	Subsystem Impact
21:30	SAS Drive Hot-Pull (TC-01/04)	Kernel logs `Attached SCSI disk [sdn]`	`osd.3` down due to stale `dm-7` mapping
22:05	10G Interconnect Outage Test	10G DAC ping timeout (192.168.10.x)	Ceph MON marks OSDs down after 20s grace
22:20	HA Maintenance Mode Activated	`lrm prox23 (maintenance mode)`	UI displays Blue Wrench icon; `pvestatd` paused
22:25	Systemd Reboot Sequence	`pam_nologin` active; transaction lock	Node status transitions to `unknown` (?)
22:29	Nodes Boot Complete & Corosync Quorum	3/3 Votes Quorate (`Ring ID: 1.10f`)	Corosync & PVE cluster sync restored
22:30	Ceph Bluestore Direct AIO Active	12/12 OSDs UP & IN; 33 PGs clean	Ceph returns to **`HEALTH_OK`** ; UI Green

4. Final Health Verification Matrix

Component	Target Metric	Verified Post-Test State	Result
Node Status	`prox22` , `prox23` , `prox24`	🟢 All 3 nodes `online` in `/cluster/resources`	**PASS**
Corosync Quorum	3 / 3 Expected Votes	🟢 3 Votes (`Quorate: Yes` , `Total votes: 3`)	**PASS**
Ceph Health	Cluster Status	🟢 **`HEALTH_OK`** (0 degraded objects, 0 down PGs)	**PASS**
Ceph OSDs	12 OSDs across 3 Nodes	🟢 **12 / 12 UP & IN** (33/33 active+clean)	**PASS**
Ceph MONs	3 MON Daemons	🟢 **3 MONs in Quorum** (`prox22` , `prox23` , `prox24`)	**PASS**
ZFS Storage	`zfs-ha` RAIDZ2 Pool	🟢 **127 TiB ONLINE** (0 read/write/cksum errors)	**PASS**
10G Network	Interconnect Latency	🟢 **0.08ms – 0.14ms (0% packet drop)**	**PASS**
HA Manager	CRM / LRM State	🟢 Quorum OK, CRM Watchdog Armed & Active	**PASS**

Component	Target Metric	Verified Post-Test State	Result
Data Loss	Object Integrity	🟢 **0 Bytes Lost / 100% Data Integrity**	**PASS**

5. Standard Operating Procedures (SOP)

SOP-A: Resolving Stale Ceph LVM Mappings After Drive Hot-Swap Without Reboot

If a drive is hot-swapped and throws `(5) Input/output error`:

```
# 1. Stop OSD target
systemctl stop ceph-osd.target

# 2. Deactivate stale LVM mappings
vgchange -an $(vgs -o vg_name --noheadings | grep ceph | tr "\n" " ") 2>/dev/null || true

# 3. Rescan SCSI bus
udevadm trigger && udevadm settle

# 4. Reactivate Ceph Volume Groups
vgchange -ay

# 5. Clear systemd burst limits and activate OSDs
systemctl reset-failed
ceph-volume lvm activate --all
systemctl restart ceph-osd.target
```

SOP-B: Clearing HA Maintenance Mode & Question Mark UI Icons

If a node shows a blue wrench or question mark after maintenance:

```
# 1. Disable maintenance mode
ha-manager node-maintenance disable <node_name>

# 2. Ensure cluster metric services are active
systemctl start pvestatd pvedaemon pveproxy

# 3. Clear any leftover migration locks on VMs
for vmid in $(qm list | awk 'NR>1 {print $1}'); do
    qm unlock $vmid 2>/dev/null || true
done
```

6. Conclusion & Production Readiness

Zero Hardware Failure: All 36x 20TB Enterprise SAS hard drives, 10G SFP+ DAC cables, and Supermicro/Dell server backplanes are in 100% operational condition.

Fault Tolerance Validated: Ceph 3x replication and ZFS RAIDZ2 successfully protected all virtual machine data through single drive pulls, multi-drive mobility swaps, 10G storage network blackouts, and node reboots.

Current Operational State: PRODUCTION READY (100% HEALTH_OK).